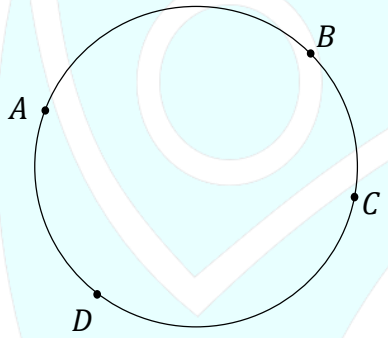
BMBA ORIGINAL SAVOLLARI

2025-yil 3-variant

1. Asosi kvadrat bo'lgan parallelipiped hovozga kafel o'rnatish kerak. Yon tomonlari kvadrati 100 ming so'mlik, asosiga esa 300 ming so'mlik kafel o'rnatilmoqda. Agar hovozning hajmi $12\,m^3$ bo'lsa, hovozning balandligi qanday bo'lganda eng kam summa sarflanadi?

A) 2
B) 4
C) 3
D) 6

2. Uchlari A, B, C va D nuqtalarda bo'lgan nechta uchburchak mavjud?



A) 6
B) 5
C) 4
D) 3

3. $\int \sqrt{e^x-1}\,dx$ aniqmas integralni hisoblang.

A) $2\sqrt{e^x-1}-2\,\text{arcctg}\,\sqrt{e^x-1}+C$
B) $2\sqrt{e^x-1}-2\,\text{arctg}\,\sqrt{e^x-1}+C$
C) $2\sqrt{e^x-1}+2\,\text{arctg}\,\sqrt{e^x-1}+C$
D) $2\sqrt{e^x-1}+2\,\text{arcctg}\,\sqrt{e^x-1}+C$

4. 16 yildan keyin otaning yoshi o‘g‘lining yoshidan 2 marta katta bo‘ladi. 4 yil oldin esa otaning yoshi o‘g‘lining yoshidan 6 marta katta bo‘lgan bo‘lsa. Ota va o‘g‘lining yoshlari orasidagi farqni toping.

A) 25
B) 26
C) 23
D) 24

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5. a ning qanday eng kichik qiymatida $a \cdot \sqrt{12} \cdot \sqrt{15} \cdot \sqrt{20} \cdot \sqrt{24}$ ning natural son bo'ladi?

A) $\sqrt{6}$
B) $\sqrt{3}$
C) $\sqrt{8}$
D) $\sqrt{2}$

6. Agar $a + b = 320$, $\frac{a}{b} = 39$ tengliklar berilgan bo'lsa, $a - b$ ning qiymatini toping.

A) 299
B) 302
C) 304
D) 307

7. p, q va p^2 arifmetik progressiyaning dastlabki 3 ta hadi. p, p^2 va q geometrik progressiyaning dastlabki 3 ta hadi bo'lsa, arifmetik progressiyaning dastlabki 9 ta hadi yig'indisini toping.

A) 6
B) 7
C) 8
D) 9

8. Soddalashtiring:

$$(x + 2) \cdot (x^2 - 2x + 4) - 4$$

A) $x^3 + 4$
B) $x^3 - 4$
C) $x^3 + 8$
D) $x^3 - 8$

9. Tenglamaning eng kichik musbat ildizi bilan eng katta manfiy ildizlari yig'indisini toping.

$$(\cos 4x + 1)(2 \sin 2x - \sqrt{2}) = 0$$

A) $\frac{\pi}{8}$
B) $-\frac{\pi}{8}$
C) $\frac{\pi}{4}$
D) $-\frac{\pi}{4}$

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10. Soddalashtiring:

$$a(b-c)-b(c-a)-c(a-b)$$

A) $2ab+2ac$
B) $2b-2ac$
C) $4ab-2ac$
D) $2ab-2ac$

11. $\begin{cases} x^2+y=2 \\ y^2+x=2 \end{cases}$ tenglamalar sistemasi nechta yechimga ega?
A) 2 ta
B) 1 ta
C) 4 ta
D) 3 ta

12. Muntazam to'rtburchakli kesik piramidaning ichiga shar ichki chizilgan. Kichik asosini yuzasi 16 ga, shar sirtining yuzasi 24π ga teng bo'lsa, katta asos yuzasini toping.
A) 24
B) 48
C) 36
D) 42

13. Kichik asosining yuzi 16 ga teng bo'lgan muntazam to'rtburchakli piramidaga radiusi 4 ga teng bo'lgan shar ichki chizilgan. Piramida katta asosining yuzini toping.
A) 256
B) 343
C) 180
D) $120\sqrt{3}$

14. $\sin 3x+\sin^3 x=\frac{3\sqrt{3}}{4}\sin 2x$ tenglamaning $\left[-\frac{\pi}{2};\frac{\pi}{2}\right]$ oraliqdagi yechimlar soni nechta?
A) $4\ ta$
B) $2\ ta$
C) $3\ ta$
D) $5\ ta$

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15. Aniqmas integralni hisoblang:

$$\int \frac{1}{x^2-4x+5}dx$$

A) $\text{arctg}(x+2)+C$
B) $\text{arctg}(x-2)+C$
C) $\text{arcctg}(x-2)+C$
D) $\text{arcctg}(x+2)+C$

16. $\frac{5}{8}-\frac{45}{88}+\frac{445}{888}-\frac{4445}{8888}$ ning qiymati qaysi oraliqga tegishli?
A) $(0;1)$
B) $(1;2)$
C) $(2;3)$
D) $(4;5)$

17. Aylananing ikkita vatari kesishadi va birinchi vatar 12 va 9, ikkinchi vatar 16 va x kesmalarga bo'linsa, x ni toping.
A) $\frac{27}{4}$
B) $\frac{28}{3}$
C) $\frac{34}{11}$
D) $\frac{18}{11}$

18. Raqamlari ko'paytmasi 18 ga teng bo'lgan nechta uch xonali sonlar bor?
A) 14
B) 19
C) 15
D) 22

19. 25 gacha bo'lgan tub sonlar yig'indisini toping.
A) 114
B) 100
C) 144
D) 99

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20. Tengsizlikni yeching:

$$log_5 \frac{4-x}{x-1} > 0$$

A) $(2;4)$
B) $(1;2{,}5)$
C) $(1{,}5;8)$
D) $(1{,}1;4{,}9)$

21. Aniqmas integralni hisoblang:

$$\int \frac{dx}{x^2-6x+10} dx$$

A) $\text{arctg}(x-3)+C$
B) $\text{arctg}(x+3)+C$
C) $\text{arcctg}(x-3)+C$
D) $\text{arcctg}(x+3)+C$

22. Soddalashtiring:

$$\frac{\cos 34^0 \cdot \cos 42^0 - \cos 56^0 \cdot \cos 48^0}{\cos 53^0 \cdot \cos 23^0 - \cos 67^0 \cdot \cos 37^0} - 1$$

A) $\sin 14^0$
B) $\cos 46^0$
C) 0
D) 1

23. Tenglamani yeching:

$$2^{x+1} + 2^{x-1} = 5$$

A) $\frac{1}{5}; 1$
B) 1
C) 0
D) 4

24. $700 \cdot 4^{13} \cdot 5^{25}$ ko'paytma nechta nol bilan tugaydi?
A) 26
B) 25
C) 27
D) 29

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25. Soddalashtiring:

$$\frac{x+2}{x-2}:\left(\frac{2x}{x^2+2x+4}+\frac{6x}{x^3-8}-\frac{1}{x-2}\right)-\frac{3(x+2)}{x-2}$$

A) $x+1$
B) $x+2$
C) x^2
D) x^3+4

26. 30 dan 50 gacha bo'lgan tub sonlar yig'indisini toping.
A) 160
B) 199
C) 94
D) 200

27. $f(x)=\frac{\ln x+1}{\ln x-1}$ funksiyaning $x_0=e^{-2}$ nuqtadagi hosilasini toping.
A) $-\frac{2e^2}{3}$
B) $\frac{2e^2}{9}$
C) $\frac{7e^2}{9}$
D) $\frac{11e^2}{9}$

28. Agar $f(x)=\frac{-2x+1}{x+4}$ bo'lsa, $\int_0^1 f^{-1}(x)dx$ ni hisoblang. (Bu yerda $f^{-1}(x)-f(x)$ funksiyaning teskari funksiyasi.
A) $9\cdot\ln 1{,}5-4$
B) $5\cdot\ln 2{,}5+4$
C) $7\cdot\ln 3{,}5-1$
D) $11\cdot\ln 4+1$

29. ABC uchburchakning AC tomoniga o'tkazilgan BD kesma, asosini A uchidan boshlab hisoblaganda 13 va 12 ga teng kesmalarga ajratadi. S_{ABD} uchburchak yuzasi 135 ga teng bo'lsa, S_{BDC} uchburchak yuzini toping.
A) $\frac{1620}{13}$
B) $\frac{1680}{13}$
C) $\frac{2245}{11}$
D) 16

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30. Quyidagi tenglamalar sistemasi berilgan: $\begin{cases}(x^2+1)(y^2+1)=10\\(x+y)(x-y)=3\end{cases}$. Tenglamaning yechimlari $(x_1;y_1),(x_2;y_2),\dots,(x_n;y_n)$ bo'lsa, $x_1+y_1+x_2+y_2+\dots+x_n+y_n$ ning qiymatini toping.

A) 5
B) 0
C) 1
D) 7

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